

SHASHWAT KUMAR

Data Scientist - II, Credit Risk Modeling at Wells Fargo & Co

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SUMMARY

Senior Data Scientist specializing in quantitative risk modeling, machine learning, and model deployment in financial services. Expert in Python, SQL, NLP and MLOps, delivering measurable uplifts in model accuracy, insights, and portfolio performance.

SKILLS

Languages Python, Apache Spark, SQL, SAS, C

Libraries Pandas, Scikit Learn, Tensorflow, Transformers

Business Monitoring, Reporting, Team Management

ML Regression, Random Forest, Boosting, NLP

AI Neural Networks, Attention Models, LLM, RAG

Tools Excel, Tableau, Docker, GCP, AWS, MLOps

EXPERIENCE

Wells Fargo & Company

Jun 2022 – Present

Data Scientist - II (Model Development Team)

Bangalore

Basel Auto Model Suite PD Recalibration through NLP Integration

- Led Probability of Default model redevelopment for Auto Indirect Loans with **\$51B** in balances across **5M accounts**
- Recalibrated **3 segments** using Logistic and XGBoost Regression, uplifting KS by 3% and MAPE reduction to 4.2%
- Constructed a feature selection pipeline using **IV, Correlation, and VIF** metrics to get **100 variables** from **1K** attributes
- Engineered an NLP pipeline leveraging **spaCy, NLTK** and **transformer models (RoBERTa)** for advanced pre-processing of **12.5M+** customer feedback records
- Embedded input sentences and **vectorized sentiment signals** to boost **AUC by 6.2%**, early default detection by **9.8%**
- Implemented a **14-grade** Consumer Risk Grade (CRG) framework using Decision Tree Splitting
- Performed statistical tests on the CRGs, achieving **\$280M reduction** in Risk-Weighted Assets across the portfolio

Auto Loan Loss PD Model Redevelopment and Implementation

- Conducted root cause analysis to identify risk factors for justifying a **20% under-prediction** of default balances
- Designed new predictive features, resulting in a **25% point** better model fit and **22% point** improvement in PD accuracy
- **Deployed** an A/B testing framework to benchmark model iterations, achieving a **12% uplift** in prediction
- Modularized performance evaluation scripts for output KPIs (PSI, VSI, KS, and GINI), cutting monitoring TAT by **30%**
- Automated CI/CD pipelines using **Docker, Airflow**, and **MLOps**, reducing model deployment time by **40%**.

Auto Vintage-Pooled Level Roll Rate Transition Model Optimization

- Revamped the 2-stage enhancement pipeline for the Roll Rate model across **6 delinquency segments**
- Augmented the loss model over the delinquency forecasting, decreasing early stage loss predictions MAPE by **47%**
- Generated and tested **20+** affordability-based roll rate variants across the transition states
- Interacted MEV features with loan-level variables to refine the monthly forecast alignment by **36%**
- Enhanced **loss rate modules** with roll-rate adjustments, improving cumulative 12-month forecast accuracy by **\$11.7M**

Dynamic VaR Engine for Multi-Asset Market Risk Management

- Built a multi-asset **Value at Risk model** for equity, fixed income, and FX portfolios, uplifting risk estimation by **28%**
- Refined tail-risk and Expected Shortfall (ES) via Monte Carlo simulations, reducing backtesting breaches by **22%**
- Created **MLOps CI/CD pipelines**, lowering deployment time by **35%** and improving risk reporting speed by **25%**

ZS Associates

May 2021 – Jul 2021

Data Science Associate Intern

Bangalore

Profitability Analysis using Insurance Sales Data

- Developed ensemble learning based models (XGBoost, Random Forest) to predict sales for **6000+ representatives**
- Achieved **0.71 F1 Score** on the validation set and **ROC-AUC of 0.84** in predicting agent sales
- Conducted clustering analysis (K-Means, Hierarchical, DBSCAN) and **t-SNE visualization** on activity and sales data
- Validated an **82%** increase in sales and a **51%** rise in activity using Hopkins-Statistic Test
- Applied feature interpretability techniques **LIME** and **SHAP** to identify high-impact sales drivers
- Forecasted a potential market opportunity worth **\$4.1M**, through targeted agent activity

PROJECTS

VGSoM, IIT Kharagpur

Aug 2021 – Dec 2021

Construction of an infrastructure financing portfolio

- Applied Markowitz mean-variance optimization on 20-year data from 100+ firms, boosting risk-adjusted returns by 12%

EDUCATION

Indian Institute of Technology, Kharagpur

Jul 2017 – May 2022

B. Tech in Industrial & Systems Engineering — M. Tech in Industrial Engineering & Management

CGPA : 8.5/10